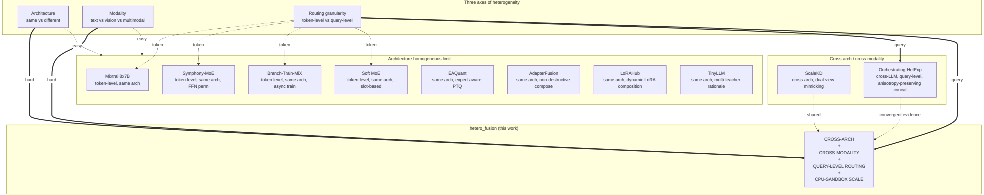
Three axes of heterogeneity
Architecture
same vs different
Modality
text vs vision vs multimodal
Routing granularity
token-level vs query-level
easy
easy
token
token
token
token
query
hard
hard
query
Architecture-homogeneous limit
Mixtral 8x7B
token-level, same arch
Symphony-MoE
token-level, same arch,
FFN perm
Branch-Train-MiX
token-level, same arch,
async train
Soft MoE
token-level, same arch,
slot-based
EAQuant
same arch, expert-aware
PTQ
AdapterFusion
same arch, non-destructive
compose
LoRAHub
same arch, dynamic LoRA
composition
TinyLLM
same arch, multi-teacher
rationale
Cross-arch / cross-modality
ScaleKD
cross-arch, dual-view
mimicking
Orchestrating-HetExp
cross-LLM, query-level,
anisotropy-preserving
concat
shared
convergent evidence
hetero_fusion (this work)
CROSS-ARCH
+
CROSS-MODALITY
+
QUERY-LEVEL ROUTING
+
CPU-SANDBOX SCALE